COMPILER
PART A (SYLLABUS)

1. Language Processing System, Compiler and Its Role, Structure of a Compiler, Phases of Compilation.
2. Lexical Analysis, Lexical Analyzer and Its Role, Tokens, Lexemes, and Patterns, Token Recognition and Token Counting, Recognition of Tokens Using RE, NFA, DFA.
3. Input Buffering, Transition Diagram, Transition Diagrams for: Keywords and Reserved Words, Identifiers, Arithmetic Operators, Relational Operators, Logical Operators, Other Common Tokens.
4. Introduction to Grammars, Context-Free Grammars (CFG), Terminals and Non-terminals, Production Rules, Derivations, Leftmost Derivation, Rightmost Derivation
5. **Syntax Analysis, Role and Responsibilities of the Syntax Analyzer, Phases of Syntax Analysis, Parsing and Its Role in Compiler Design, Top-Down Parsing, Recursive-Descent Parsing, Left Recursion, Left-Recursive Grammars, Direct and Indirect Left Recursion, Why Left Recursion Is a Problem for Recursive-Descent Parsers, Detection of Left Recursion, Elimination of Left Recursion, Transformation of Left-Recursive Grammars into Equivalent Non-Left-Recursive Grammars**
6. Lexical Errors and Their Detection, Syntax Errors and Their Detection, Common Causes of Lexical and Syntax Errors.